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Enforcement of Emissions Trading

Sanction regimes of greenhouse gas emissions trading in the EU and China

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Enforcement of Emissions Trading

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Abstract

This chapter aims to further the debate regarding the role of law for establishing an adequate enforcement strategy for an emissions trading scheme. We focus on sanction regimes within the EU ETS and the Chinese emissions trading pilot projects. Section 2 sets the scene by pointing at the need of an adequate enforcement approach and related legal scholarship. Section three presents the specific case of the EU ETS, established with Directive 2003/87/EC from 2003, while section 4 turns to the recently developed greenhouse gas emissions trading pilot projects in China. Particularly the sections 3 and 4 focus on sanctions for excess emissions, thereby discussing recent case law regarding penalties for emissions trading within the EU and specific enforcement approaches in the emerging emissions trading regimes in China. Section 5 concludes, thereby highlighting that proper evaluations of compliance with emissions trading regimes, is a challenge in itself, not only for governments, but also for academics who want to gain further insights into how emissions trading regimes work in the practice of different legal systems. In conclusion, access to information regarding compliance and enforcement with an emissions trading regime needs to be further explored, not only for China but also for the EU.

Key words:

Emissions trading, excess emissions penalty, access to information, EU ETS, Chinese emissions trading pilot projects.

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1. Introduction

This contribution aims to provide new insights into the role of law for establishing and maintaining a well-operating emissions trading scheme for which an adequate enforcement strategy is essential.¹ Since emissions trading is an instrument for environmental policy, compliance of participants with the emissions trading scheme is of imminent importance in order to avoid unacceptable damage to the environment. Indeed, while the government may decide on the level of environmental protection by determining an emissions cap (which in fact determines the level of allowable pollution under the emissions trading regime), proper compliance is needed to ensure that this environmental target will not be superseded. Consequently, a trustworthy *measurement* of emissions has to take place, together with a submission of a number of tradable allowances exactly corresponding to the amount of emissions an operator has produced. In addition, *sanctions* need to be established in order to ensure that

1) unlawful action – meaning action against the rules of the emissions trading regime - will be deterred, and

2) in case unlawful emissions took place (meaning emissions not covered with a corresponding emission allowance submitted to the government), these so called “excess emissions” will be compensated by the infringing operator so that finally the environmental target will be reached.

A one-size-fits all blueprint for the monitoring and sanction rules for an emissions trading regime doesn't exist, although some convergence of approaches among jurisdictions may happen. A range of different design options can be developed, which may also depend on the legal system in which an emissions trading regime will operate. When discussing how monitoring and enforcement has to be set up within an emissions trading regime, a balance

¹ This contribution (finished in March 2015) builds on several publications of the authors, among which Marjan Peeters: Enforcement of the EU greenhouse gas emissions trading scheme, in: K. Deketelaere, M. Peeters: EU Climate Change Policy: The Challenge of New Regulatory Initiatives, Edward Elgar, 2006, p. 169-187; Inspection and market-based regulation through emissions trading: the striking reliance on self-monitoring, self-reporting and verification, Utrecht Law Review, Volume 2, Issue 1, June 2006; The Enforcement of Greenhouse Gas Emissions Trading in Europe: Reliability Ensured? in: Lee Paddock, Du Qun, Louis Kotze, David L. Markell, Kenneth J. Markowitz, DurwoodZaelk, Compliance and Enforcement in Environmental Law: Toward More Effective Implementation, Edward Elgar 2011, 407-430. Huizhen Chen: Inspection and Enforcement in Chinese Carbon Emissions Trading: Progress, Problems, and Prospect, Environmental Law Reporter, Vol. 44, No. 7, July 2014, pp.10596-10606; and also: Towards a market-based climate regime in China? A legal perspective on the design and implementation of greenhouse gas emissions trading (forthcoming PhD thesis at Maastricht University).

has to be found between on the one hand developing an effective approach that ensures that infringements and particularly excess emissions will be avoided, and, on the other hand, taking account of the legal rights private actors enjoy. Meanwhile, examination of practice may furthermore contribute to the further understanding of how to enforce emissions trading, since in a number of jurisdictions the instrument has been applied for several pollution problems.

Remarkably, legal scholarship has thus far paid limited attention to the enforcement dimension of emissions trading. This chapter aims to further the debate but since the topic would merit a book instead of a chapter, a choice is made to concentrate on sanction regimes. Section 2 first sets the scene by pointing at the need of an adequate enforcement approach and related legal scholarship. Then section three will present the specific case of the EU ETS, established with Directive 2003/87/EC from 2003, while section 4 turns to the recently developed greenhouse gas emissions trading pilot projects in China. Particularly the sections 3 and 4 focus on sanctions for excess emissions, thereby pointing at recent case law regarding penalties for emissions trading within the EU and at specific enforcement approaches in the emerging emissions trading regimes in China. Section 5 concludes.

2. Legal and economic perspectives on enforcement of emissions trading

2.1 The need for an adequate sanction regime

As a starting point, we assume that the price of the allowances not only creates an incentive to attain environmental protection at low cost, but at the same time may serve as an incentive for firms to avoid compliance with the rules.² This “double edged sword” is particularly a risk in the absence of an effective enforcement system.³ How to set up an enforcement approach has been considered to some extent in economic literature, particularly also by Tietenberg who has manifested himself as an expert on the emissions trading instrument.⁴ From an economic point of view, the emitters — as rational economic persons — generally make up decisions of compliance or non-compliance by weighing the costs and benefits of each choice, in order to

² Peeters (2006) p. 179; also Epiney (2012) p. 27.

³ OECD (2004), p.25.

⁴ Tietenberg (2006), p. 171-176, particularly p. 171.

minimise their own costs.⁵ Economic literature argues that cost-minimizing emitters would choose a level of compliance at which the marginal cost of compliance is equal to the expected marginal cost of noncompliance, which is defined as the likelihood that a non-compliance will be detected and a sanction levied, multiplied by the marginal sanctions (such as the fines).⁶ In this respect, economists propose that regulators can raise the expected cost of non-compliance by paying attention to: a) the likelihood that violations will be detected and punished, and b) the level of the sanctions.⁷ In this context, it should be ensured that the expected cost of non-compliance such as the fine per unit of excess emissions is (sufficiently) higher than the allowance's price.⁸ The two conditions mentioned above should not only stimulate the operator to submit truthful emissions reports, but also guarantee that each operator will hold sufficient allowances to cover his emissions.⁹

In order to reduce and to remove the incentive for non-compliance, the instrument of a *financial penalty* imposed on offenders with excess emissions is widely considered to be a necessary and effective approach to be included into an emissions trading enforcement regime. However, the financial penalty may be ineffective if it exceeds the financial capacity of the offenders and thus additional approaches and penalties would be necessary.¹⁰ Particularly the compensation of the excess emissions may become a problem when the company goes bankrupt. Also the possibility to impose criminal sanctions for specific noncompliance including false reporting and other illegal conduct is suggested.¹¹ It is generally believed that when optimal dissuasion can be achieved equally through either fines or criminal sanctions, fines are preferred as they are less expensive to administer.¹²

⁵ *Ibid*, p.171.

⁶ *Ibid*.

⁷ *Ibid*, p.173.

⁸ Stranlund *e.a.* (2002), p.346.

⁹ Tietenberg (2006), p. 175.

¹⁰ Becker (1968); Shavell (1985), referred to by Peeters (2011), p. 421. An evaluation report of the EU ETS mentions that in Germany several cases of insolvent operators have been the case, see Verschuuren, Fleurke (2014) p. 33.

¹¹ Tietenberg (2006), p.182.

¹² See Polinsky, Shavell (1979), referred to by Faure (2010), p.266.

2.2 States as participants to an emissions trading regime

The need for an adequate enforcement regime also applies in the situation where *states* are the participants to an emissions trading scheme, which is for instance the case with the Kyoto Protocol (article 17) and the EU Effort Sharing Scheme (Decision 406/2009/EC). Within the latter EU scheme, emission reduction targets are addressed to Member States. In view of reaching those targets transfer of emission reduction achievements may happen among states, next to other flexibility options like the use of Clean Development Mechanism credits (CERs).¹³ To what extent Member States will make use of the trading options remains to be seen: the compliance period runs from 2013 until 2020. In order to let the trading operate efficiently, it is of imminent importance that the Commission will start infringement actions against Member States that have not reached their annual emission reduction obligations, otherwise potential non-complying states may be less interested into using trading opportunities in order to cover the shortfall of their target. However, the possibilities for keeping Member States in compliance with EU law are not strong; the infringement procedure that the European Commission may undertake for holding Member States to account is not effective enough.¹⁴

On a more global scale, the issue of how to ensure that states comply with emission trading rules is a concern of the Kyoto Protocol, which includes the project-based mechanisms known as Joint Implementation (JI) and CDM next to interstate trading of so-called Assigned Amount Units (AAUs). The enforcement branch of the Compliance Committee to the Kyoto Protocol has some means to supervise proper compliance with the set emission reduction obligations.¹⁵ The number of parties with emission reduction targets under the Kyoto Protocol (including the Doha amendment from 2012) is limited and mainly consists of EU Member States. Nonetheless, experiences with the Kyoto Protocol Compliance Committee are useful for gaining further understanding of what is needed and what kind of strategies are suitable for holding states in an international emissions trading regime compliant. Also the EU,

¹³ See for a more elaborated discussion Peeters, Stallworthy (2012).

¹⁴ Peeters (2013) section 3 (compliance).

¹⁵ See about the specific features this compliance committee for instance Loibl (2010) and Oberthür, Lefebvre (2010).

particularly its experience with the Effort Sharing Decision, provides a source for further insights on how states as participants to an emissions trading regime will act and how compliance can be reached. Next to considerations of the behaviour of EU Member States as participants to an emissions trading regime, a proper implementation by Member States of the EU ETS – which covers emissions by industries - is crucial for ensuring that no excess emissions take place. Within the EU ETS, Member States are primarily responsible for monitoring and enforcing the behaviour of the covered industries, and the European Commission may start infringement procedures if Member States fall short regarding this. Part of the problem is that the Commission may not always have sufficient information on possible lacks in implementation and / or execution by Member State.¹⁶ Another issue regarding compliance by states arises if emission trading regimes are linked between several countries. This is for instance the case for Norway, Iceland and Lichtenstein that have joined the EU ETS. In order to protect the integrity of the EU ETS, it is of course necessary that sufficient guarantees are established to ensure that these three countries apply an adequate monitoring and enforcement approach.¹⁷

2.3 A need for more legal scholarship on enforcement of emissions trading

This section will shortly review the limited appearance of legal literature concerning enforcement of emissions trading, particularly regarding the EU ETS. As such, quite some legal literature has emerged after the introduction of greenhouse gas emissions trading in the EU in 2003. The bibliography in a book that critically reviews the development of legal scholarship on the EU ETS covers 21 pages.¹⁸ A review of this list, together with other English legal literature that is known to us, confirms that the question of how to ensure compliance within the EU ETS has thus far got limited attention.¹⁹ We can however point at two relatively recent important publications:

¹⁶ Epiney (2012) p. 27.

¹⁷ As far as is known to us, this issue has not yet been discussed in depth in legal literature.

¹⁸ Bogojević (2013).

¹⁹ We tried to get a comprehensive overview and are grateful to Dionysios Stivas, student fellow at the Metro institute of Maastricht University in 2013–2014, who did a literature search in spring 2013. See for an interesting comparative overview Kruger, Egenhofer (2006). Since the implementation of the enforcement provisions of the EU ETS has to be done by Member States, it may be that in the languages of these countries interesting case law and legal literature has emerged.

- Pablo Mendes de Leon (2012) has discussed the legality of the sanctions applicable to aviation emissions covered by the EU ETS. This article takes a firm position against the inclusion of international aviation emissions. The author argues that the EU, including the Court of Justice of the EU, has acted disrespectful of international law, more specifically the international agreement regarding civil aviation.²⁰ Particularly the legality of the operating ban, a sanction of the EU ETS system that is provided only for aircraft operators, may be at risk according to this scholar.

- With regard to (the much needed) empirical studies, a study, commissioned by the EU, was published in 2014. This study, developed by Verschuuren and Fleurke, discusses experiences from 6 Member States with monitoring and enforcement of the EU ETS. The study (further discussed in section 3.3) finds *inter alia* that EU ETS compliance practice in the 6 investigated Member States varies greatly.²¹ This illustrates the need for additional comparative research discussing different enforcement approaches (also those from other EU member states), including the need for considering further harmonisation of national enforcement practice.

3. Sanction Regime of the EU ETS

3.1 The legislative system of the EU ETS

Achieving full compliance with the EU ETS depends on, firstly, the activities by the covered entities, secondly, the due performance of the verifiers, and, thirdly, the enforcement strategies of the Member States and the three states that have joined the EU ETS. The EU ETS relies heavily on self-monitoring and self-reporting by covered entities, backed up by a third party verification of the compiled report.²² The reliance on private verifiers means that competition may take place among these verifiers in order to win contracts by the covered entities; this adds to the market-based regulatory character of the emissions trading instrument. The idea of using private actors for compliance control is not necessarily limited to the EU ETS but can also be considered for command and control approaches. The use of

²⁰ Mendes de Leon (2012). See case C-366/10 Air Transport Association of America and Others v Secretary of State for Energy and Climate Change, Judgment of the Court (Grand Chamber) of 21 December 2011 (not yet report).

²¹ Verschuuren, Fleurke (2014) p. 2 (executive summary).

²² Peeters (2006b).

private verifiers means that a system has to be set up to control the verifiers, and to ensure that fraud by verifiers can be effectively addressed. While admittedly much more research is needed to understand in depth how the combination of self-monitoring, self-reporting and the use of third party verifiers works, and whether this strategy is to be preferred above governmental control, this chapter focuses on the sanctions regime since regarding this issue even less legal literature has been published.²³ This section aims to provide some modest input to this much needed scholarly attention.²⁴

The EU ETS Directive prescribes Member States, firstly, to establish and implement a national legislative framework with penalties regarding infringements of national provisions needed to implement the Directive.²⁵ As a general rule, it is provided that these penalties must be effective, proportionate and dissuasive. This instruction leaves some discretion to Member States as to how to design the national legislative framework. Next to this, the directive prescribes two specific sanctions.²⁶ These sanctions relate to the core rule that no emissions may happen without a corresponding surrendering of emission rights.²⁷ It concerns the following sanctions:

1. The imposition of a financial penalty, called the “excess emissions penalty”.²⁸ The level of this penalty is prescribed in the Directive, meaning that it shall be EUR 100 for each tonne of carbon dioxide equivalent emitted for which the operator (including aircraft operators) has not surrendered allowances.²⁹ From 1 January 2013 onwards the level of the excess emissions penalty shall increase in accordance with the European index of consumer prices.³⁰ Payment of the excess emissions penalty shall not release the operator from the obligation to surrender an amount of allowances equal to those excess emissions, which may be done when surrendering allowances in relation to the following calendar year.³¹

²³ Epiney discusses the (limited) possibilities for individuals to enforce correct compliance by Member States in the EU (p. 27-29).

²⁴ Within the EU ETS, several criminal activities have been employed like particularly VAT fraud and money laundering. These activities as such, most likely, do not threaten the environmental integrity of the scheme which consists of the idea that no emissions may happen without allowances. See for a court decision related to the case of stolen allowances and liability of the EU (claim rejected): T-317/12 (18 September 2014).

²⁵ Art. 16(1) EU ETS directive.

²⁶ Art. 16(2) EU ETS directive (naming and shaming) and art. 16(3) (financial penalty).

²⁷ This rule is established in article 12(3) EU ETS directive. See furthermore also art. 15 EU ETS directive establishing that in case an emissions report has not been verified as satisfactory cannot make further transfers of allowances.

²⁸ Art. 16(2) EU ETS directive.

²⁹ See about a potential legal barrier for Member States to impose a more severe penalty Jans, Vedder (2012) p. 118.

³⁰ Art. 16(4) EU ETS directive.

³¹ Art. 16(3) EU ETS directive.

2. On top of the excess emissions penalty, Member States shall ensure publication of the names of operators (including aircraft operators) who are in breach of requirements to surrender sufficient allowances.³² There is no instruction how this publication has to be done. Moreover, the publication is only needed in the language(s) of the specific Member States (and hence, for instance, in the specific official journal of the Member State) which may be a barrier for easy insight into the names of offenders.

Next to this, and specifically for air craft operators, Member States *may request* the Commission to impose an operating ban to an air craft operator for which the national enforcement measures have not resulted in compliance.³³ The fact that it is the Commission, and not a national authority, that imposes such a ban means that access to the court for air craft operators that want to fight against such a ban will be on the EU level. All other legal actions by industries against Member States' enforcement decisions have to be submitted to the national courts.

3.2 Case law on the EU ETS

In the meantime, case law has shown that the CJEU favours a strict enforcement approach regarding the excess emissions penalty.³⁴ The case concerned a failure by two Swedish industries to surrender sufficient allowances before the deadline of 30 April, and as a consequence the excess emissions penalty was imposed.³⁵ The industry however held that it was due to an error, and not unwillingness, to surrender before the deadline, stating that it had sufficient allowances on its account but failed to surrender them. The referring Swedish court wanted to know whether the financial penalty has to be applied irrespective of the cause of the omission, and, secondly, whether there may be a possibility to waive or reduce the penalty in case of an administrative error or a technical problem while the operator had a sufficient

³² See for a critical discussion Peeters (2006a) p 180-181. In depth research is needed for getting a better understanding of the usefulness and acceptability of the naming and shaming approach. One can also wonder whether information about compliance by the operators should already be publicly available, upon request, in view of Directive 2003/4/EC.

³³ Art. 16(5) EU ETS directive. Mendes de Leon (2012) states that not imposing this sanction on stationary installations is disputable in view of the principle of equal treatment, p. 297. He furthermore argues that exercise of this sanction by the Commission on aviation companies from third countries (and not by an administering Member State with which a third country has treaty relationships) might be illegal, p. 301.

³⁴ CJEU C-203/12 (17 October 2013).

³⁵ C-203/12, para 17 and 18: "As at 30 April 2007, the Billerud companies, companies governed by Swedish law holding carbon dioxide emission allowances, had not surrendered the allowances equal to their emissions for 2006 (10 828 and 42 433 tonnes respectively). Consequently, the Naturvårdsverket imposed the penalty provided for by Law No 2004:1109 implementing Directive 2003/87, in the amount of SEK 3. 959.366 for one company and SEK 15.516.051 for the other (EUR 433.120 and EUR 1.697.320)."

amount of allowances. The Court considers that Article 16(3) and 16(4) ³⁶ “*must be interpreted as precluding operators who have not surrendered, by 30 April of the current year, the carbon dioxide equivalent allowances equal to their emissions for the preceding year, from avoiding the imposition of a penalty for the excess emissions for which it provides, even where they hold a sufficient number of allowances on that date*”. Interestingly, the Advocate General favoured a more flexible approach, based on the reasoning that given the specific circumstances of the case (an administrative error by the operators) the excess emissions penalty does not apply, but the more general rule laid down in article 16(1) EU ETS Directive obliging Member States to provide adequate penalties.³⁷ Accordingly, one could also consider that in order to avoid administrative mistakes (that can lead to huge financial consequences for the operators) administrative authorities could establish a system of warnings in order to ensure that operators are aware of their duty to surrender their allowances before the deadline.³⁸ Basically, the Court chooses to follow a strict interpretation leaving no room to take into consideration the circumstances that led to the breach. In theory, the application of emissions trading does not necessarily require such a strict enforcement approach, but incorporating the possibility to consider the proportionality of the sanction of course may imply more governmental capacity.³⁹ In the specific EU context, moreover, discretion in defining the level of the penalty embodies the danger of sparing the industry.⁴⁰ In the meantime, one can see that on the national level some exceptions are provided to the application of the excess emissions penalty. Germany allows an exemption in case of force majeure.⁴¹ UK legislation provides that the excess emissions penalty will be reduced to 20 euro if an operator who initially failed to surrender the correct allowances, but advised the regulator of their mistake and surrendered the correct allowances before the regulator noted their noncompliance.⁴² The question of legality of these national measures may be submitted to the courts.

³⁶ As applicable by the time of the offence.

³⁷ C-203/12, Opinion of Advocate General Mengozzi, delivered on 16 May 2013, para 33.

³⁸ Peeters (2014). For instance in Poland, a national authority sends reminders before each important deadline: Verschuuren, Fleurke (2014) p. 61.

³⁹ Peeters (2014).

⁴⁰ Peeters (2006a) p. 182.

⁴¹ Verschuuren, Fleurke (2014) p. 32.

⁴² Verschuuren, Fleurke (2014), p. 68, see for the UK text <http://www.legislation.gov.uk/ukxi/2012/3038/regulation/54/made>.

Another question for preliminary ruling has been submitted to the CJEU in February 2014, and the Court has yet to provide judgment.⁴³ While in the Swedish case discussed above the possibility of mitigating or waiving the financial penalty is being discussed, this case concerns the question of whether the financial penalty has to be applied when the emissions report was approved by the verifier but later is found by the public authority to be incorrect in the sense of showing an insufficient amount of emissions.⁴⁴ The question submitted for preliminary ruling reads as follows: “*Must Article 16(3) and (4) of Directive 2003/87/EC be interpreted as meaning that the excess emissions penalty must also be applied in the case where an operator has, by 30 April of a given year, surrendered a number of allowances corresponding to the total emissions stated in its report on emissions from the installation for the preceding year, and that report has been assessed as satisfactory by the verifier, but where the competent authority, after 30 April, has established that the verified emissions report had erred by understating the total quantity of emissions, the report was duly corrected and the operator surrendered the additional allowances within the new period for surrender?*”

The German Federal Administrative Court who has referred this question to the CJEU elaborates the potential positive or negative answer to the question, and seems to be in favour of a negative answer, both in view of the text of the Directive and the principle of proportionality. The referring decision considers *inter alia* that the Directive does not prescribe that the excess emissions penalty has to be applied when after approval of the verifier a governmental authority deviates from this decision.⁴⁵

3.3 Discovering non-compliance in the EU ETS

Epiney (2012) has stressed some potential enforcement problems with the EU ETS. She particularly states that the Commission may not always have sufficient information on possible lacks in implementation and / or execution by Member States.⁴⁶ Also researchers face dilemmas when trying to understand the level of compliance of operators with the EU ETS. Naturally, they cannot do investigations regarding the behaviour of industries themselves, and are hence dependent on information provided by industries or by national

⁴³ C-148/14; see for the decision of the German Federal Administrative Court (Bundesverwaltungsgericht) <http://www.bverwg.de/entscheidungen/entscheidung.php?ent=200214B7C37.11.0>.

⁴⁴ German Federal Administrative Court Decision, para. 4.

⁴⁵ *Ibid.*, para 23.

⁴⁶ Epiney (2012) p. 27, see also Peeters (2012) p. 423-424.

authorities. Obviously, operators and national authorities may have reason to stay reluctant with providing comprehensive and correct information, since disclosure of noncompliance by operators, or disclosure of short falling enforcement strategies by Member States, may have consequences in the form of negative publicity and sanctions, including infringement actions by the Commission towards Member States. Any evaluation research by academics regarding compliance with the EU ETS (and other environmental legislation) hence meets quite serious barriers for finding the truth and may mostly be based on indirect information. The evaluation report regarding the effectiveness of the compliance mechanism of the EU ETS, published in 2014, is for instance largely based on information from existing written sources and interviews with key players in the compliance mechanism (particularly national authorities from 6 Member States). Based on these materials, the report states that compliance with the EU ETS is very high.⁴⁷

The report provides a string of more detailed and interesting insights into how compliance with the EU ETS works in practice. One finding is for instance that the number of staff employed in the national emissions authorities differs enormously;⁴⁸ this may give rise to assumptions that in countries with low capacity some non-compliance will stay undetected. Moreover, and strikingly, the researchers have found that in one of the six Member States (Hungary) there was “unsatisfactory access to the necessary Hungarian sources”.⁴⁹ The researchers also point at the fact that the information provided by Member States in reports pursuant to article 21 of the EU ETS directive do not provide a complete picture regarding actual compliance and enforcement of the EU ETS in the Member States. These findings illustrate the barriers to getting full insight into the rate of compliance by operators with the EU ETS.

Regarding naming and shaming, the report concludes that this sanction is not actively applied in the 6 researched Member States. Two interesting facts are been reported:

- The names of installations that did not surrender sufficient allowances are far from easy to find.⁵⁰
- German practice shows that, at least according to the competent authority, NGOs do not yet follow up on the naming and shaming information.⁵¹ This could be explained

⁴⁷ Verschuuren, Fleurke (2014) p. 78 para 6.

⁴⁸ *Ibid*, p. 78

⁴⁹ *Ibid*, p. 76.

⁵⁰ *Ibid*, p. 79.

by the fact that most of the detected infringements are not qualified as deliberate fraud actions, but are merely in the sphere of mistakes due to the complexity of the rules.⁵²

These short examples show that more research is needed towards the usefulness and application of naming and shaming. If a legislator wants to use naming and shaming in order to apply any deterrent effect, there should be provisions that the names will be known to those who for instance might want to buy allowances (or the products, since consumers may want to boycott industries that do not obey the climate law rules). However, it remains to be seen how deterrent this information disclosure is, and whether it will be used by ENGO's.

The report furthermore discloses that Member States provide a range of different administrative and criminal sanctions. The authors suggest some further harmonization, for instance by broadening the scope of the Directive 2008/99/EC on the protection of the environment through criminal law to the EU ETS.⁵³ An important forward look given by the evaluation report is that enforcement challenges may become more serious if the price of the allowances increases.⁵⁴ The compliance mechanism of the EU ETS will then be put at the test.

4. Sanction regime of the Chinese CO₂ ETS

China has set up a national carbon intensity reduction target binding the Chinese governments in the 12th Five-Year Planning (2011-2015), intending to reduce national carbon dioxide (CO₂) emissions by 17% per unit of GDP, compared to that of 2005.⁵⁵ In order to facilitate the achievement of this target, seven pilot projects for CO₂ ETS, with the approval of the National Development and Reform Committee (NDRC), are being established and implemented in five municipalities and two provinces.⁵⁶ These pilots take place in view of a possible national ETS in the future.⁵⁷ The enforcement packages established within these sub-national ETS pilots contain various penalties for non-compliant behaviour, primarily related to causing excess emissions and infringing MRV-rules. In addition, a target-based accountability system

⁵¹ *Ibid*, p. 37.

⁵² *Ibid*, p. 78.

⁵³ *Ibid*, p. 78.

⁵⁴ *Ibid*, p. 33-34.

⁵⁵ Chapter 3, Guidelines of the Twelfth Five-Year Planning for National Economic and Social Development [hereafter referred as the 12th Five Year Planning (2011-2015)], National People's Congress, 16 March 2011 (in Chinese)

⁵⁶ Five municipalities are Beijing, Shanghai, Tianjin, Shenzhen and Chongqing, and the two provinces are Guangdong and Hubei.

⁵⁷ NDRC, The Notice on Initiating Pilot Programs of Emissions Trading, 29 October 2011 (in Chinese).

regulating local governments and governmental officials aims to stimulate proper governmental enforcement action.

4.1 Regulatory frameworks establishing sanctions in China

This section will introduce first the sanction regimes of the pilot projects, with a focus on penalties for excess emissions, after which the specific Chinese governmental accountability system will be presented. A discussion of these two approaches takes place in section 4.2 and 4.3.

*Penalties for excess emissions*⁵⁸

Each pilot project has established its own enforcement regime, including sanction regimes for excess emissions. The discussion below takes the Shenzhen pilot project as a starting point⁵⁹, supplemented by an explanation of major differences in other pilot projects.

For excess emissions in the Shenzhen pilot project, the competent authority, the Shenzhen Development and Reform Committee (DRC), shall order the emitter to surrender allowances to cover the excess emissions within a specific time period or will deduct the allowances directly from the holding account, including, if necessary, from next year's holdings.⁶⁰ In addition, a financial penalty in an amount equal to three times the average market price of the last six continuous months will be imposed on the emitter with excess emissions.⁶¹ Moreover, causing excess emissions also invokes other penalties, including:

- “naming and shaming” by putting the noncompliance into the credit record⁶² of the offender and disclosing it to the public;

⁵⁸ The pilot projects also prescribe various sanctions for other noncompliance behaviour. The focus of this chapter goes to penalties for excess emissions.

⁵⁹ The reason mainly relies on the fact that Shenzhen pilot is the first CO₂ ETS pilot started, and, moreover, relatively detailed provisions regarding enforcement have been provided in this pilot.

⁶⁰ Article 75, Interim Measures on Carbon Emissions Trading Management in Shenzhen, Shenzhen Municipal People's Government, 19 March 2014 (in Chinese).

⁶¹ *Ibid*; see also Article 8, Provisions of Controlling Carbon Emissions for Shenzhen Special Economic Zone, Shenzhen People's Congress Standing Committee, 10 November 2012 (in Chinese).

⁶² The credit record of an enterprise contains its basic registration information, commercial credit information, and other relevant information that may affect the enterprise's credit, such as the illegal behaviour record that has been punished by the government. See Article 43 of Administrative Measures of Shenzhen Municipality on Credit Information Collecting and Credit Rank Evaluating of Enterprise, 19 November 2002 (in Chinese).

- the disqualification in applying for relevant governmental financial funding for five years;
- the notification of the non-compliance record of a State Owned Enterprise (SOE) to the municipal state-owned assets supervision and administration commission and including it into the performance evaluation system for the SOE (including its leading cadres).⁶³

Similar sanctions for excess emissions have been established by most of the other pilot projects but with different designs and intensities. For instance, twice the number of the shortfall of allowances will be deducted from the allowances of next year's holding by the provincial DRC in the pilot projects of Guangdong and Hubei.⁶⁴ In contrast, other pilot projects such as Shanghai and Tianjin only provide that the governmental authorities would 'order the emitters to make the correction' without any specificity as to how.

Imposing a financial penalty for excess emissions is a standard option in most pilot projects, but the level of the penalty and the way how it is determined differs. The fine for the excess emissions in Beijing ranges from three to five times the average market price of the allowances.⁶⁵ However, in other pilots the financial penalties have not been directly linked to the market price and the level of the fines differs: the excess emissions will result in a fixed fine of 50,000 Chinese Yuan (CNY) in Guangdong⁶⁶ while the fines range from 50,000 to 100,000 CNY in Shanghai⁶⁷. In the Hubei pilot project, the fine ranges from one to three times the market price for the excess emissions, with a total maximum of 150,000 CNY.⁶⁸

Target-based accountability system for local governments/cadres

⁶³ See Article 65, Interim Measures on Carbon Emissions Trading Management in Shenzhen

⁶⁴ Article 37, Trial Measures of Guangdong Province on Carbon Emissions Management, Guangdong Provincial Government, 15 January 2013; see also Article 46, Interim Measures of Hubei Province on the Management and Trading of the Carbon Emission Right, Hubei Provincial government, 4 April 2014 (in Chinese)

⁶⁵ Decision on Implementation of Carbon Emissions Trading Pilot on Premise of Strict Total Control of Carbon Emissions, Beijing People's Congress Standing Committee, 27 December 2013 (in Chinese)

⁶⁶ Article 37, Trial Measures of Guangdong Province on Carbon Emissions Management

⁶⁷ Article 39, Trial Management Measures for Carbon Emissions in Shanghai, Shanghai Municipal People's Government, 18 November 2013 (in Chinese)

⁶⁸ Article 46, Interim Measures of Hubei Province on the Management and Trading of the Carbon Emission Right

The pilot projects to be managed by the selected local governments are aimed to contribute to the achievement of sub-national carbon intensity reduction targets that have been determined for these local governments. At the same time, these local governments are subjected to a target-based accountability system, holding the leading cadres of local governments responsible for the fulfilment of these targets.

According to the 12th Five-Year Planning (2011-2015), the results of the performance evaluation concerning the fulfilment of the carbon intensity reduction target, including the sub-national targets, will provide an important basis for the election, promotion, punishment and awards for the leading cadres.⁶⁹ In this respect, the NDRC has formulated specific performance assessment measures regarding the implementation and completion of the distributed carbon intensity reduction targets at the provincial levels.⁷⁰ Similar to the accountability system applied in the domains of economic development and social management, the leading cadres may be punished by disciplinary sanctions including warning, demotion, or even dismissal once any failure of a specific target achievement is detected.⁷¹ By means of the incorporation of the target achievement into the cadre system, the Five-Year Planning has significant binding effect on the local governments, and has in this respect a potential powerful effect.⁷²

Furthermore, according to the amendment of the Environmental Protection Law in 2014, a target-based responsibility system and performance assessment system for environmental protection will be implemented in the future, requiring the government to incorporate the completion of environmental protection targets, as an important basis for assessment of the competent departments and persons in charge thereof.⁷³ This amendment suggests the linkage between the political accountability mechanism and the legal system, but the exact design and detailed implementation of this new regime is not yet further explained. While this regime may stimulate the local governments to apply an ETS with stringent sanctions on excess emissions that may impede the fulfilment of the carbon intensity reduction target, it deserves

⁶⁹ Section 3, Chapter 61, 12th Five-Year Planning (2011-2015).

⁷⁰ The Notice of the NDRC on Printing and Distributing the Performance Assessment Measures Regarding the Carbon Intensity Reduction Target Responsibility, NDRC, 6 August 2014 (in Chinese)

⁷¹ This is usually called “One-vote veto” policy (*yi piao fou jue zhi*) in China, since these targets have the “veto power”, the failure to achieve which can trump other accomplishments. See for instance Deng (2011).

⁷² Lin (2012), p.11; see also Mol, Carter (2006), p.157.

⁷³ Article 26, Environmental Protection Law of China, promulgated by Standing Committee of the National People's Congress, 26 December 1989, revised 24 April 2014, effective as of 01 January 2015

further discussion in order to understand to what extent it will contribute to the effective implementation of the pilot projects, particularly also the imposition of excess emissions penalties.

4.2 Comparison and discussion of the sanction regimes in China

Legal basis for penalties in the pilot projects

In order to ensure compliance and instil confidence in the ETS, it is necessary to establish a binding and sufficiently stringent sanction regime. The legal basis and specific implementation procedures for penalties to be applied by local governments in the pilot projects should obviously be consistent with the Chinese legal system. Because most of the penalties that would be imposed by the pilot projects will be administrative penalties, the institution and implementation of sanctions for noncompliance ought to be designed in accordance with the Administrative Penalty Law of China to ensure that the sanctions will be enforceable.⁷⁴ Administrative penalties may be created only through formal legislations including laws, administrative regulations, local regulations, and specific rules of ministries and local governments.⁷⁵ In light of the fact that greenhouse gases are still not regulated as pollutants in China, the sanctions provisions established in the Environmental Protection Law can't be applied directly to the CO₂ ETS pilot projects. Until the end of November 2014, most of the regulatory documents relating to the pilot projects are issued in the form of 'other normative documents' (*qi ta gui fan xing wen jian*) adopted by local governments that do not constitute formal legislation in China. Important exceptions are the local regulations adopted by the Shenzhen and Beijing authorities, and four rules of local governments promulgated for the pilot projects of Shanghai, Guangdong, Shenzhen and Hubei.⁷⁶ Hence, one may raise questions as to whether the existing regulatory frameworks established in the pilot projects, in particular those that have not yet adopted local legislation, have sufficient legal status to impose specific penalties.

⁷⁴ Article 3, Administrative Penalty Law of China, promulgated by National People's Congress, 17 March 1996

⁷⁵ *Ibid*, at Article 14.

⁷⁶ According to the Chinese legislative structure, the provincial people's congresses and their standing committees may enact local regulations, while the provincial government can adopt rule of local government in accordance with laws; the legal effect of the local regulations is higher than that of the rules formulated by local governments at and below the corresponding level. See Article 63, 73, 80, Legislation Law of China, promulgated by National People's Congress, 15 March 2000, revised 15 March 2015 (in Chinese)

The Administrative Penalty Law as well as the Legislative Law in China permits the imposition of most administrative penalties through local regulations, with the exception of restriction on personal freedom and withdrawal of a business license of an enterprise.⁷⁷ However, there are limitations provided by the Administrative Penalty Law as to the type of administrative penalties that may be imposed under the rules of certain local governments; for example, the administrative penalty of a disciplinary warning or certain level of fine may be all that is permitted when no laws or regulations have been enacted.⁷⁸ Therefore, some of the administrative penalties necessary for the ETS could be implemented through local regulations and rules of local government adopted in the pilot projects. However, criminal liability for noncompliance can only be established at the national level.

Fines (like the excess emissions penalty) can be created by law, administrative regulation, or local regulation.⁷⁹ In addition, a certain level of fine could be imposed based on the rules of ministries and provincial governments.⁸⁰ In this respect, the Standing Committees of the People's Congress of Shenzhen and Beijing as well as the municipal / provincial governments of Shanghai, Shenzhen, Guangdong and Hubei have the authority to establish a financial penalty for noncompliance through local regulations and rules of local government. At the same time, the absence of local legislation is also the main reason for the lack of financial penalties in the pilot projects of Tianjin and Chongqing, which have only adopted the governmental normative documents rather than relevant local legislation yet. Hence, in line with the Chinese legislative system, there is not much legal barrier for provincial (and certain municipal) authorities, such as the local governments implementing the pilot projects, to impose a fine on excess emissions through local regulation and rule of local government.

Effectiveness of the financial penalty

⁷⁷ Article 11, Administrative Penalty Law of China

⁷⁸ *Ibid*, at Article 13.

⁷⁹ *Ibid*, at arts. 9-11.

⁸⁰ *Ibid*, at arts. 12-13.

The financial penalty, particularly the excess emissions penalty, is assumed to play a key role in ensuring compliance of the ETS-participants.⁸¹ However, one may question whether the excess emissions penalties established in the Chinese pilot projects are stringent enough.

In the pilot projects of Shenzhen and Beijing, financial penalties for excess emissions established in the local regulations are indexed to the prevailing market price. Because the amount is prescribed, the local authority does not need to determine the specific amount of the fine, apart from gathering data on the prevailing market price. Meanwhile, imposing a fine equal to three (or up to five) times the market price on the violator, demonstrates the government's attempt to raise the cost of breaking the law. Emitters, as rational economic players, might be expected to choose to buy allowances to cover excess emissions from the market at market prices rather than paying three (or five) times as much in fines.

In contrast, the Shanghai and Guangdong pilot projects permit the competent authorities, usually the provincial DRC, to determine the fine within a certain range stipulated in the rules of local government. As for the Hubei pilot project, the fine is connected to the market price but limited to a cap of 150,000 CNY.⁸² It is still an open question whether the penalties applied in the pilot projects, such as a cap on the penalty of 100,000 CNY in Shanghai, will serve as an effective deterrent. Especially when the cost in reducing emissions or buying allowances is higher than that of the capped fine, emitters may simply choose to pay the fine. However, the local governments have quite limited legislative competence to create the financial penalties, the specific maximum amount of which shall be laid down by the standing committees of the provincial people's congresses.⁸³ In this respect, for instance, the maximum for the fine that can be established by Hubei provincial government has already been raised largely from 30,000 to 150,000 CNY by the standing committee of the people's congresses of Hubei province⁸⁴, although it may remain relatively low for some big industries.

Although the revised Environmental Protection Law cannot provide a solid and direct legal basis for the pilot projects as mentioned above, the strict sanctions for illegal discharge of

⁸¹ Cheng (2013), p.32

⁸² The "cap" seems to be the maximum amount of fine for excess emissions, even when the price of the allowance is very high, the excess emission is very large, or when the operator is a relatively small company.

⁸³ See Article 13, Administrative Penalty Law of China

⁸⁴ See The Regulation of the standing committee of the people's congresses of Hubei province on the maximum amount of the administrative fine established by governmental rule, adopted on 21 September 1996, revised on 23 May 2013 (in Chinese).

pollutants established in the amendment to the Environmental Protection Law in 2014 may pave the way for a stringent financial penalty punishing the excess emissions covered by the ETS, if specific GHG is explicitly labelled and regulated as a pollutant in the future. To be specific, if an emitter is fined and ordered to make a correction due to the illegal discharge of pollutants but refuses to make such a correction, the competent authority may impose the fine again on a daily basis consecutively, equal to the original amount of the fine multiplying the days commencing from the date immediately following the date when it is ordered to make a correction.⁸⁵ There will be no cap on such a fine until the correction is done by the emitter. Moreover, it is stipulated that local regulations may increase the types of illegal acts subject to the fine consecutively on a daily basis in light of the actual needs of environmental protection.⁸⁶ In this sense, there seems to be a possibility for the local ETS pilot projects to impose this strict fine on the excess emissions in relevant local regulations.

Implementing the enforcement measures

Because the excess emissions penalty is deemed to be an administrative penalty, the decision-making has to be in conformity with the Administrative Penalty Law, which requires the administrative body to conduct an investigation in a comprehensive, objective, and fair manner and to collect relevant evidence.⁸⁷ However, the detailed provisions of these procedures are still missing in some of the local regulations or other relevant rules in the pilot projects to date.

Some other challenges may occur in the pilot projects. For instance, the Beijing DRC attempts to limit the discretion for imposing administrative penalties by adopting detailed standards for decision-making, in the form of an “other normative document” instead of formal legislation, but whether this will turn out effective remains to be seen.⁸⁸ On the basis of the fact, seriousness and relevant factors of the illegal act, the noncompliance is categorised into four groups subjected to (1) impunity, (2) lighter punishment, (3) normal punishment and (4) heavier punishment. Moreover, the specific legal basis, types and ranges of the penalties for noncompliance emerging in the Beijing pilot have been stipulated in a set of trial executive

⁸⁵ Article 59, Environmental Protection Law of China

⁸⁶ *Ibid.*

⁸⁷ Article 36, Administrative Penalty Law of China

⁸⁸ Decision on the regulation of the discretion in implementing the administrative penalties in the carbon emissions trading, Beijing DRC, 6 May 2014 (in Chinese)

standards in the form of a detailed table.⁸⁹ This decision together with the executive standard is supposed to control the exertion of the administrative discretion.⁹⁰ Because of its legal status, however, this decision and the standard can only be considered as a kind of policy provision. Nonetheless, this provision appears to be a self-regulatory document binding the Beijing DRC *per se*, aiming to reduce and to avoid abuse of discretion. It will be important to observe whether and how the application of such standards for administrative discretion (*cai liang ji zun*) lead to effective and proportional deterrents in the sanction regime for the Chinese ETS.⁹¹

Furthermore, under the Administrative Penalty Law administrative counterparts, including citizens, legal persons, and organizations, have the right to refuse to accept an administrative penalty and may apply for administrative reconsideration or bring an administrative lawsuit to appeal the penalty.⁹² Emitters participating in emissions trading have the same right to appeal the administrative decision in court, although it remains questionable whether such case law will emerge in China. If that is the case, the case law will provide a source with further information on how the sanction regime can be applied in practice and what legal protection ETS participants may get. Especially in view of the powerful government but weak judicial system in China, in particular the fact that the courts are funded by the governments at the same level, the limited role of judicial system and independent court decision is a concern.⁹³ Given the currently quite limited role of courts in China, it would be important to ensure the private actors can challenge the government's sanction decision before courts, if China indeed wants to follow the rule of law.

4.3 Concluding remarks

In addition to the legal enforcement provisions in the ETS pilots, non-legal mechanisms will be used for achieving a reliable ETS. The penalties for the participants' non-compliance elaborated above have shown that the sanctions for the SOEs are connected with the cadre system. In view of the fact that ETS pilot projects are meant to contribute to compliance with

⁸⁹ For instance, with regard to the financial penalties for excess emissions ranging from three to five times of the average market price, the lighter punishment will be three times, while the normal punishment will be four times and the heavier punishment will be five times. *Ibid.*

⁹⁰ *Ibid.*, at Article 1,4

⁹¹ For more on this topic Yu (2008); Zhou (2007) (Chinese literature).

⁹² Article 6, Administrative Penalty Law of China

⁹³ Du (2009), p.441-442

the carbon intensity reduction target, the cadre responsibility system is expected to have a significant influence on the leaders of the local government, stimulating them to apply an effective ETS pilot project with stringent punishment on excess emissions that may endanger the achievement of the carbon intensity reduction target. In this sense, the cadre system remains an important approach, controlling the local governments and SOEs in view of ensuring well-functioning ETS pilot projects.

However, it remains questionable to what extent the enforcement package of the pilot projects can ensure compliance. The pilot projects are encouraged to explore feasible approaches to emissions trading, including the establishment of an effective enforcement package. It is inevitable that diversity of specific enforcement rules developed by local authorities seeking to accommodate different local circumstances will develop. However, the effectiveness of a sanction regime will largely depend on the legislative competence of the local authorities, and in this respect there are limits concerning the types and intensity of the sanctions regime established in the pilot projects. Furthermore, the question remains whether this decentralised approach for sanctions by local authorities will lead to problems of a “race-to-bottom” among the regions in which the ETS projects are applied. This could happen in order to attract more industries to stay in the local territory, aiming to ensure local economic growth in view of the Chinese GDP-oriented evaluation system. In this respect, a harmonised or unified sanction regime at the national level deserves further consideration, in particular if China wants to move to a national ETS. However, a rule regarding the establishment of a possible national ETS, adopted by the NDRC recently, indicates that the sanctions for noncompliance will be mainly designed and implemented by the provincial authorities.⁹⁴ The specific design and implementation of the decentralised approach under the national scheme needs to be observed and the lessons from the pilot projects will be relevant.

5. Conclusion

This chapter has focused on sanction regimes within the EU ETS and the Chinese emissions trading pilot projects, and has shown that a financial penalty to be imposed if excess emissions have taken place forms a core sanction in these regimes. Both in the EU and in the

⁹⁴ See Article 40-42, The Interim Measures for Management of Carbon Emissions Trading, NDRC, 10 December 2014, effective as of 10 January 2015 (in Chinese).

Chinese pilot projects, the financial penalty is backed up with additional sanctions, like naming and shaming in the EU, and exclusion from governmental funding in China.

Regarding the EU ETS, some case law is developing particularly regarding the extent to which an administrative authority may (or has to) take into consideration the circumstance which has led to the infringement by the operator. In a judgment, and contrary to the opinion of the Advocate General, the CJEU does not allow a flexible approach regarding the imposition of the excess emissions penalty (which could, in essence, be a more proportional approach). In the meantime, some Member States (Germany and the UK) have included in their national legislation some possibilities to deviate from a strict imposition of the financial penalty but such provisions are, given the Court's decision, at risk to be incompatible with EU law. Some further discussion will take place since a German court has referred a new case to the CJEU, which prolongs the discussion about how to enforce the EU ETS particularly in view of a positive assessment by the verifier of the emissions report.

Within China, a country that aims to move to the rule of law but which still suffers from the relatively weak enforcement of legislation and heavily depends on strong political steering by the central government, local governments that are selected to establish pilot emissions trading programmes face competence problems with establishing rules that would support a comprehensive enforcement, including adequate sanctions. At the same time, through the cadre system, local governmental officials can be held responsible for the performance of the carbon intensity reduction target, which then has significant impact on their specific design choice and application of the emission trading instrument. In line with the decentralised experimental approach in establishing the pilot projects, the diverse design features of the sanction regimes at the local level may however imply the risk of a 'race-to-bottom' endangering the integrity of an ETS, as far as a national scheme is concerned. Since the pilot projects are approved by the national government with a view on establishing a national ETS system, it will be interesting to see whether and how the local pilot projects will be evaluated in order to derive learning points for moving to a national scheme. Such evaluation can also shed some further light on how enforcement actually works in the specific Chinese context: on the one hand, it needs to be discussed whether the practice of imposing excess emission penalties will be effective enough, and, on the other hand, how the imposition of penalties will be done in view of the Chinese rule of law and how, in this respect, operators may have access to independent courts. One aspect that has not been discussed in this chapter,

but which deserves certainly also scholarly attention, is how Environmental Non-Governmental Organisations could play a role for holding operators compliant.⁹⁵

Taking also a forward look in the case of the EU ETS, a potential increase of the price of the allowances will put the compliance regime to the test and may provide further insight into the practice and legality of imposing sanctions on participants to the EU ETS, including the operating ban to aviation companies. At the same time, doing proper evaluations of emissions trading regimes, particularly where it concerns compliance and enforcement, is a challenge in itself, not only for governments, but also for academics who want to gain further insights into how emissions trading regimes work in the practice of different legal systems. Most likely, barriers exist to getting real insight into non-compliant behaviour. In conclusion, access to information regarding compliance and enforcement with an emissions trading regime needs to be further explored, not only for China but also for the EU.

⁹⁵ McAllister (2010).

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